

Playing Tougher for One's Size: Positional Physicality in the NHL

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*Statistics and Data Science, Computer Science
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What does CSAx mean?

CSAx measures physical presence relative to a player's listed height and weight.

DIRECT CONTACT

Hits given or taken, blocks, fights, contact penalties

PUCK + SPACE

Net-front plays and forechecking; defensive recoveries and entry denials

HIGHER CSAx More physical presence for one's size

LOWER CSAx Less physical presence for one's size

2024–25 top three

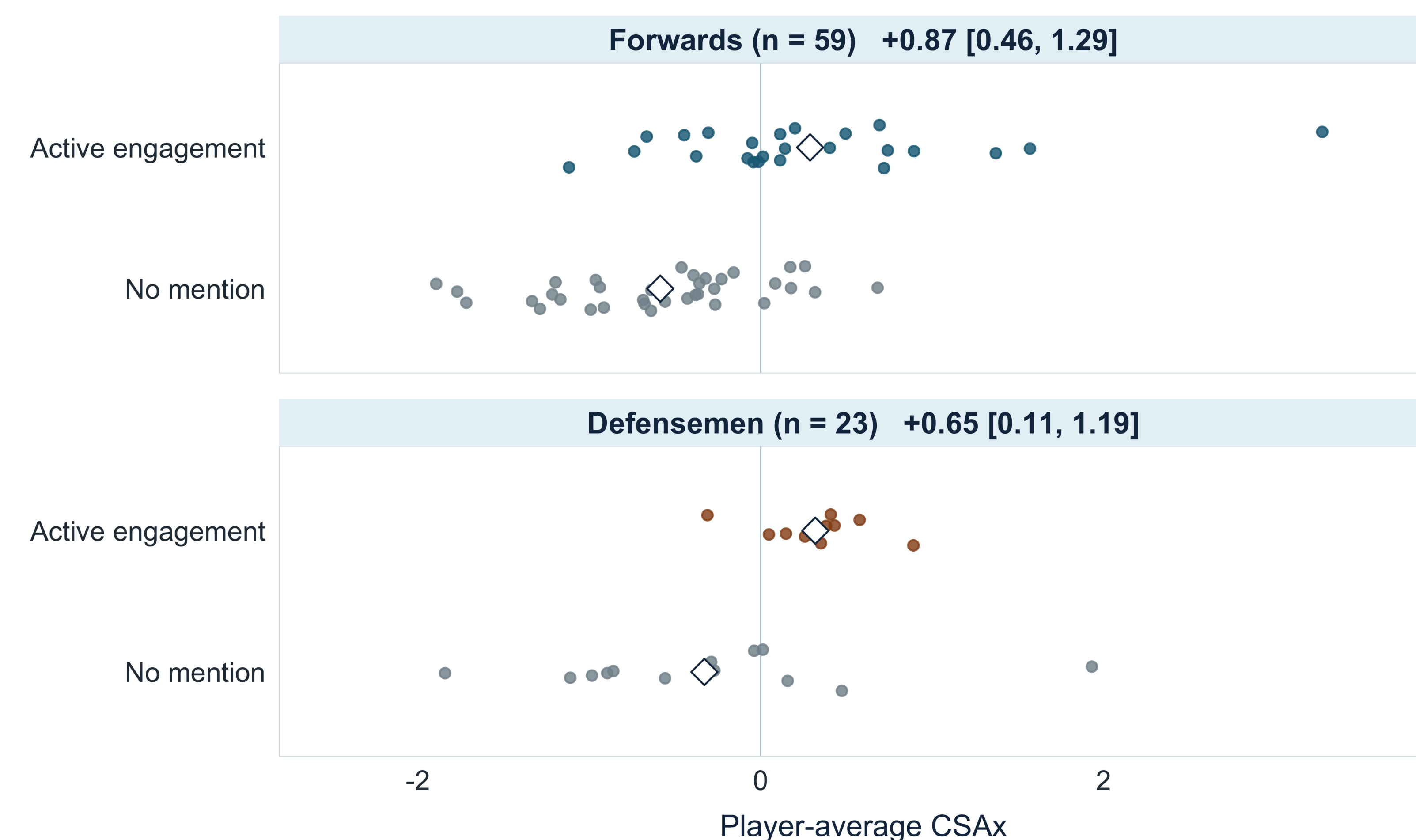
FORWARDS			DEFENSEMEN		
#	Player	CSAx	#	Player	CSAx
1	Sam Carrick	4.69	1	Connor Clifton	2.95
2	Mathieu Olivier	3.88	2	Radko Gudas	2.93
3	Zachary L'Heureux	3.28	3	Nick Seeler	2.82

Does it match what scouts see?

Will Cuyllé | 2020 NHL draft guide

“Powerful skater with good speed. ... Heavy on the body and on pursuit of pucks. Very strong puck protection and wall play. Excellent net-front presence.”

NHL Central Scouting, 2020 draft guide



Each dot is a player; diamonds mark group means. Brackets show 95% intervals. Active-engagement descriptions align with higher CSAx in both positions.

Where does it matter?

Special-teams ice time

Share of total ice time; CSAx +1 versus -1

	POWER PLAY	PENALTY KILL
Forwards	-1.2 points	+1.4 points
Defensemen	-0.9 points	+1.3 points

Playoff hitting | forwards

First four playoff games vs. regular-season baseline

Lower CSAx	+136%
Higher CSAx	+84%

Hits rise in both forward profiles, with a smaller proportional rise for higher-CSAx players; the defenseman contrast is uncertain.

Next-season roster continuation

Chance of reaching 300 NHL minutes; CSAx +1 versus -1

Forwards	+2.9 percentage points
Defensemen	+4.2 percentage points

Among returners, higher CSAx shows no clear increase in minutes per game.

Behind this study

My NHL Stats R&D internship let me draw on advice from supervisors and colleagues as this research took shape.

Next: Carnegie Mellon poster, MIT Sloan submission, and a journal manuscript.

Sources

NHL play-by-play and official ice time
All Three Zones (Corey Sznajder)
NHL Central Scouting draft guide



2025 NHL intern cohort



Full 2024–25
rankings and
technical details